TECHNICAL REPORT

ROGII Pipeline Review and Improvement Plan

Technical review of the user's ROGII submissions, local and Kaggle notebooks, public leaderboard, and community research.

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Technical summary

- **Confirmed position:** your best submitted public RMSE is **7.201**, rank **466 / 4,634** (top **10.08%**) on the July 10 snapshot. The leader is at **4.859**; top 100 is **7.035** and top 200 is **7.134**.
- **The 7.159 label is not a verified score for your current output.** A currently published notebook titled `7.091 PUBLIC` produces a `submission.csv` byte-identical to yours, yet your actual submission scored 7.201. Notebook titles and current outputs therefore cannot be used as score evidence without version-level submission history.
- **Your pipeline is technically broad but hard to learn from.** It mixes PF, beam search, NCC, spatial priors, boosted models, projection, contact overrides, and Gold calibration, while most full ablations are disabled and the saved booster/Ridge CV remains around 10.4. That makes public-LB tuning too influential.
- **Recommended direction:** freeze the 7.201 lineage as an anchor, build a multi-cut well-level validation harness, measure seed variance, and focus on candidate selection/uncertainty plus genuinely decorrelated models. Do not spend the next cycle retuning blend weights inside the same highly correlated public lineage.

7.201
YOUR BEST PUBLIC RMSE

8
SCORED SUBMISSIONS

Confirmed submission history

Source: Leaderboard and submission audit

466
CURRENT RANK

10.08%
TOP SHARE

Public leaderboard snapshot

Source: Leaderboard and submission audit

4.859
LEADER PUBLIC RMSE

2.342
GAP TO LEADER

Competitive distance

Source: Leaderboard and submission audit

7.155
APPROX. TOP-5% CUTOFF

241
TEAMS AT OR BELOW 7.159

Dense leaderboard region

Source: Leaderboard and submission audit

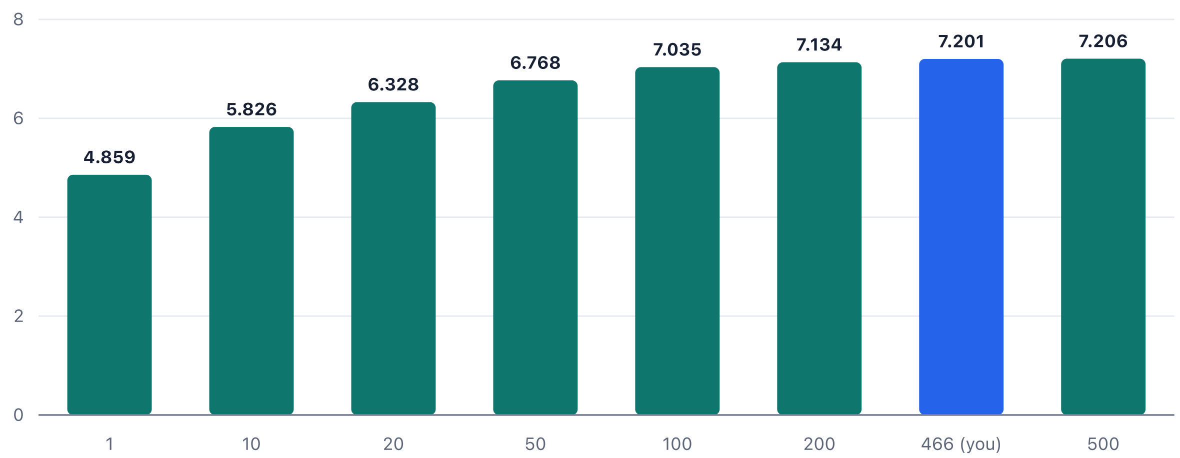
The leaderboard is extremely dense around your score

A 0.042 move from 7.201 to 7.159 corresponds to roughly rank 466 to rank 241 in this snapshot. That sensitivity makes single public submissions a poor tuning signal, especially for stochastic particle-filter pipelines.

Public leaderboard RMSE cutoffs

Source: Leaderboard cutoff audit

Public RMSE (lower is better)



Scope, data, and metric definitions

The competition metric is pooled row-level RMSE, so long and badly drifting wells dominate through squared error. The visible local test folder contains only three example wells; Code Competition scoring reruns against a hidden set of about 200 wells, with roughly one quarter contributing to public LB according to organizer/community discussion. Conclusions drawn from the three visible wells-especially overlap overrides-do not establish hidden-test performance.

The leaderboard and submission facts below are current as of July 10, 2026. Community CV/LB numbers are useful directional evidence but are not independently reproduced.

Your scored submission history

Source: Submission history audit

DATE	PUBLIC RMSE	DESCRIPTION	SUBMISSION REF
2026-07-07 19:39	7.201	--ok try public notebook	54,439,017
2026-07-03 14:09	7.252		54,300,470
2026-07-01 14:51	7.669		54,235,686
2026-06-30 21:05	7.573	--remove gold prefix calibration stage	54,212,419
2026-06-30 10:58	7.288	trying base submission	54,198,353
2026-06-29 13:20	7.271	base submission	54,170,908
2026-06-28 21:01	7.31	base	54,150,893

DATE	PUBLIC RMSE	DESCRIPTION	SUBMISSION REF
2026-06-28 19:38	16.714		54,149,747

The current notebook has breadth but insufficient experimental isolation

The submitted notebook is a 60-code-cell pipeline. Its saved outputs load three LightGBM models (RMSE 10.47-10.77), two CatBoost models (10.55-10.58), and a Ridge stack with overall RMSE 10.4197. It then builds a physics selector, runs a second Fleongg-style pipeline, projects `U = TVT + Z`, blends the branches, applies guarded contact reconstruction, and runs visible-prefix Gold calibration.

For the three visible sample wells, the contact override replaces every prediction row and the Gold profiles become numerical no-ops. That is a useful implementation check, not an estimate of hidden performance. The hidden rerun lacks train-only formation columns and mostly lacks same-well overlap, so the meaningful core remains the PF/beam/typewell alignment, test-available geometry, and any learned residual model.

External research points to selection and new signal-not another public fork

The reviewed evidence consistently separates candidate generation from candidate selection. PF, beam, HMM, formation surfaces, and typewell matching can produce very good oracle paths, but ambiguous GR makes it difficult to choose the right branch. Public notebooks also share artifacts and code lineage, producing highly correlated errors and substantial seed noise.

Evidence from reviewed approaches

Source: Research evidence audit

APPROACH	EVIDENCE	IMPLICATION	CONFIDENCE
Version-level notebook audit	Byte-identical public notebooks can vary about 0.1-0.4 RMSE; current 7.091 output equals user's 7.201 file.	Do not tune from titles or one stochastic run; archive hashes and average seeds.	High
Worst-well analysis	Repeated hard wells show 20-50 RMSE from early drift and branch persistence.	Track worst-decile SSE and add uncertainty-based fallback to conservative anchors.	High
Multi-cut HMM smoother	Public research reports PF 4.90, HMM 5.18, and 50/50 blend 4.57 on five sample wells; uncertainty-error correlation up to 0.92.	Backtest on all wells; use posterior mean/std as candidate and gating features.	Medium
Candidate-path oracle	Discussion reports oracle over PF/beam/formation candidates around 4.5-6 RMSE, while selection remains difficult.	Invest in a cross-fitted path ranker or gating model, not only new generators.	Medium
Standard tabular ensembles	Top participant reports LGBM/CatBoost/XGBoost without exotic feature engineering; input representation mattered most.	Rebuild clean test-available feature sets and auxiliary targets before changing model class.	Medium

APPROACH	EVIDENCE	IMPLICATION	CONFIDENCE
Neural sequence models	Community results range from CV 5s for strong non-tabular work to CV 8.9/LB 8.39 and unstable branch jumps.	Do not lead with a large U-Net; first prove a small ranker/smoother adds decorrelated OOF signal.	Medium

Validation design should become the center of the project

Use 5-fold GroupKFold by well, but create multiple synthetic prediction starts per training well at matched prefix fractions. Report pooled RMSE, mean per-well RMSE, worst-decile pooled RMSE, performance by eval length/Z-span/GR missingness, and seed-to-seed standard deviation. Keep all cuts from the same well in one fold. Compare every candidate on the identical cuts and preserve out-of-fold predictions for blending and selector training.

The key robustness checks are: true prediction-start evaluation versus synthetic cuts; deep-tail versus early-tail performance; deterministic versus stochastic reruns; and error correlation between candidate families. A change should ship only when it improves pooled OOF RMSE across seeds or adds enough error diversity to improve a cross-fitted blend.

Recommended experiment sequence

The ordering below is designed to answer high-value questions before spending GPU time. Each experiment has an explicit gate; if it fails, stop that branch rather than layering more heuristics.

Prioritized experiment backlog

Source: Experiment backlog

PRIORITY	EXPERIMENT	SUCCESS GATE	COST
1	Multi-cut GroupKFold harness + experiment ledger	Reproduces baseline across 5 folds; reports pooled/worst-decile RMSE and seed std.	Medium
2	Freeze and rerun core candidates deterministically	OOF predictions saved for hold, PF scales, beam, NCC, projection, tabular stack; pairwise error correlations computed.	Medium
3	Cross-fitted candidate gate / path ranker	Beats best fixed blend by ≥0.15 pooled RMSE across at least 3 seeds and does not worsen worst decile.	Medium
4	Exact HMM forward-backward candidate	Adds ≥0.05 blend gain or improves calibrated uncertainty after runtime-normalized comparison.	High
5	Test-available GBDT with auxiliary future targets	Improves pooled OOF ≥0.10 and remains stable on deep-tail and high-GR-missing segments.	Medium
6	Hard-well fallback policy	Cuts worst-decile SSE ≥10% with total pooled RMSE non-inferior.	Low

PRIORITY	EXPERIMENT	SUCCESS GATE	COST
7	Only then: compact 1D CNN / cross-attention ranker	Adds decorrelated OOF gain after ensembling; no confidence-jump failures.	High

Limitations, uncertainty, and failure modes

- Public LB is noisy: reviewed version histories show byte-identical stochastic notebooks moving roughly 0.1-0.4 RMSE. Treat a single 0.05-0.15 improvement as inconclusive.
- Public notebook titles are stale evidence. The current `7.091 PUBLIC` output is byte-identical to your 7.201 output.
- Multi-cut validation is not perfectly distribution-matched: earlier artificial cuts have shorter prefixes and different tail lengths. Report both matched-cut and true-start metrics.
- Hidden-test formation markers are absent. Train-only `ANCC/AST*/EGFD*/BUDA` fields may be used to create targets or train spatial priors, but inference features must be reproducible from `MD, X, Y, Z, GR, TVT_input` and typewell `TVT/GR`.
- Sequential methods can make an early branch error and carry it through an entire well. Worst-well diagnostics and uncertainty gating are required, not optional.

Further questions

1. Which subset rule separates CV wells that correlate with LB, as hinted by top participants-trajectory depth, GR coverage, formation regime, or prediction-zone geometry?
2. Can HMM posterior variance, PF seed dispersion, GR misfit, and candidate disagreement predict the winning tracker out of fold?
3. Which candidate family remains least correlated with the frozen 7.201 anchor after honest multi-cut evaluation?
4. Are the worst wells dominated by under-estimated structural drift, branch switching, missing GR, or typewell mismatch? The answer should determine whether the next model is a smoother, ranker, or residual learner.

Sources and provenance

Reader-facing source details retained from the validated report artifact. Local implementation paths and export-runtime metadata are intentionally omitted.

1. **Kaggle public leaderboard export**
   Kaggle CLI full public leaderboard export; selected ranks and score thresholds were computed with pandas. · Retrieved: 2026-07-10T14:36:30Z · Dataset: leaderboard_cutoffs_2026-07-10.csv
2. **User submission history from Kaggle CLI**
   Authenticated Kaggle CLI submissions listing for flexonafft. · Retrieved: 2026-07-10T14:34:00Z · Dataset: competition submissions: rogii-wellbore-geology-prediction
3. **Local and Kaggle notebook audit**
   Structured inspection of notebook cells, saved outputs, Kaggle-pulled source, and downloaded output artifacts. · Retrieved: 2026-07-10T14:45:00Z · Dataset: publicscore-note-lb-7-159.ipynb · Dataset: *.ipynb · Dataset: *
4. **Kaggle competition discussions and public notebooks**
   Read-only review of current Kaggle discussion threads and selected public notebooks. · Retrieved: 2026-07-10T14:44:00Z · Dataset: 719389 · Dataset: 717573 · Dataset: 721549 · Dataset: 718670 · Dataset: 723815 · Dataset: 723647 · Dataset: 722041 · Dataset: 722236 · Dataset: 714514
5. **Leaderboard and submission audit**
   Combine the reviewed leaderboard cutoff and submission-history CSVs. · Retrieved: 2026-07-10T14:50:00Z · Dataset: leaderboard_cutoffs_2026-07-10.csv · Dataset: submission_history_2026-07-10.csv
6. **Leaderboard cutoff audit**
   Read selected public leaderboard ranks from the reviewed snapshot. · Retrieved: 2026-07-10T14:50:00Z · Dataset: leaderboard_cutoffs_2026-07-10.csv
7. **Submission history audit**
   Read the user's completed scored submissions from the reviewed Kaggle CLI extract. · Retrieved: 2026-07-10T14:50:00Z · Dataset: submission_history_2026-07-10.csv
8. **Research evidence audit**
   Read the curated evidence table derived from version-level artifacts and attributed Kaggle discussions. · Retrieved: 2026-07-10T14:50:00Z · Dataset: research_evidence_2026-07-10.csv
9. **Experiment backlog**
   Read the evidence-backed prioritized experiment backlog. · Retrieved: 2026-07-10T14:50:00Z · Dataset: experiment_priorities_2026-07-10.csv